RECITATION

Setting up your development environment

Goal:

- Linux OS
- Cmake
- VS Code
- Hello World working

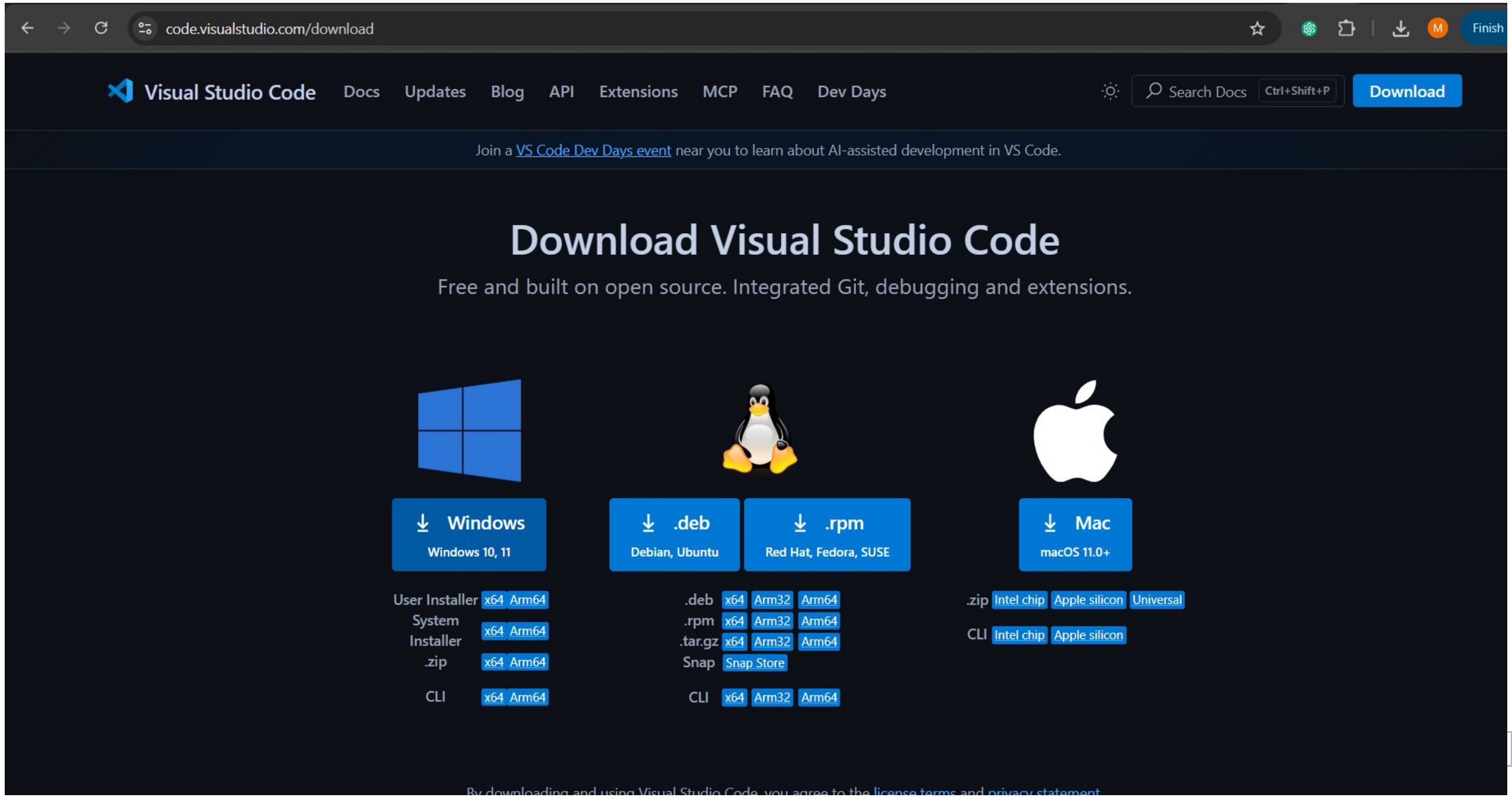
RECITATION - Router

This is for those who want to use linux

```
If you already have Linux:
        if you have VSCode:
                Go To Step 5
        else:
                Install VSCode and Go To step 5
else:
        if you have VSCode on your windows:
                Go To Step 1
        else:
                Go To Step 0
```

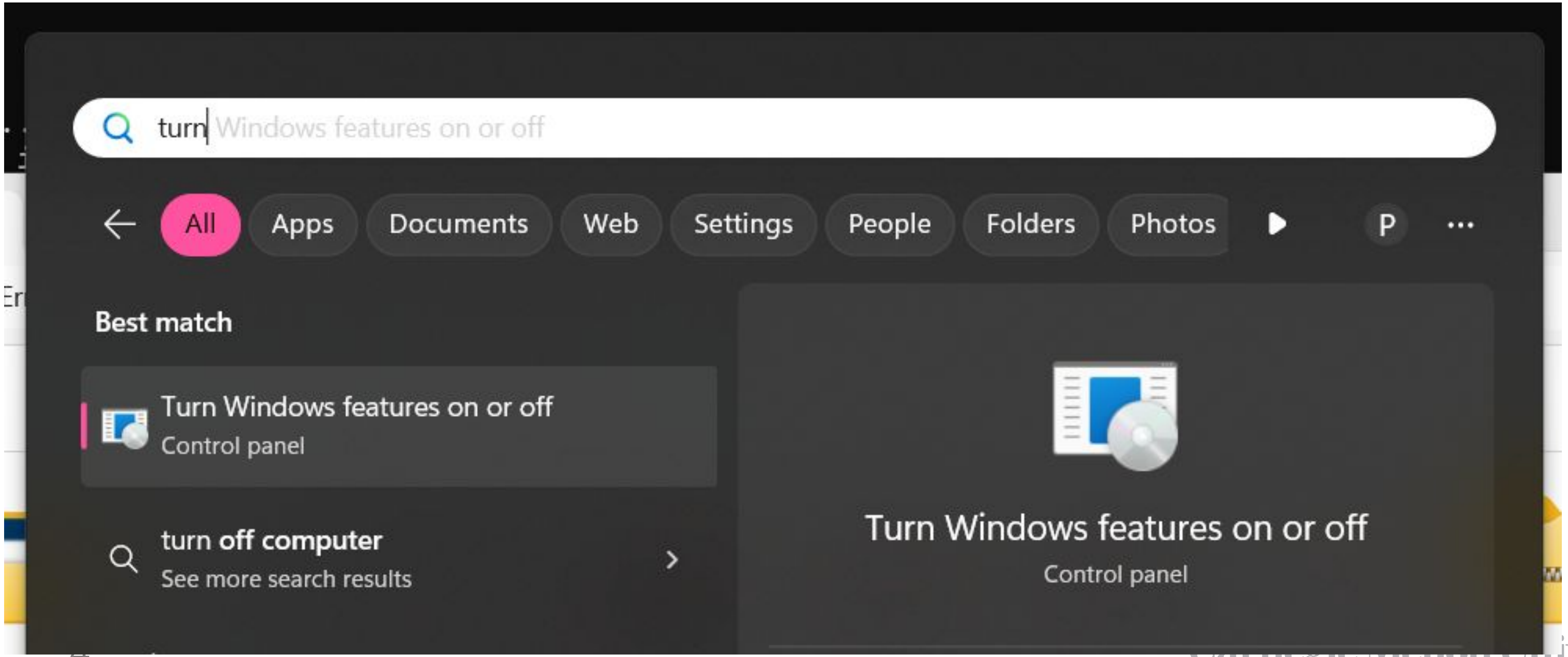
Step 0 - Download VSCode in your windows

Download VSCode from this Link (Select the windows one)



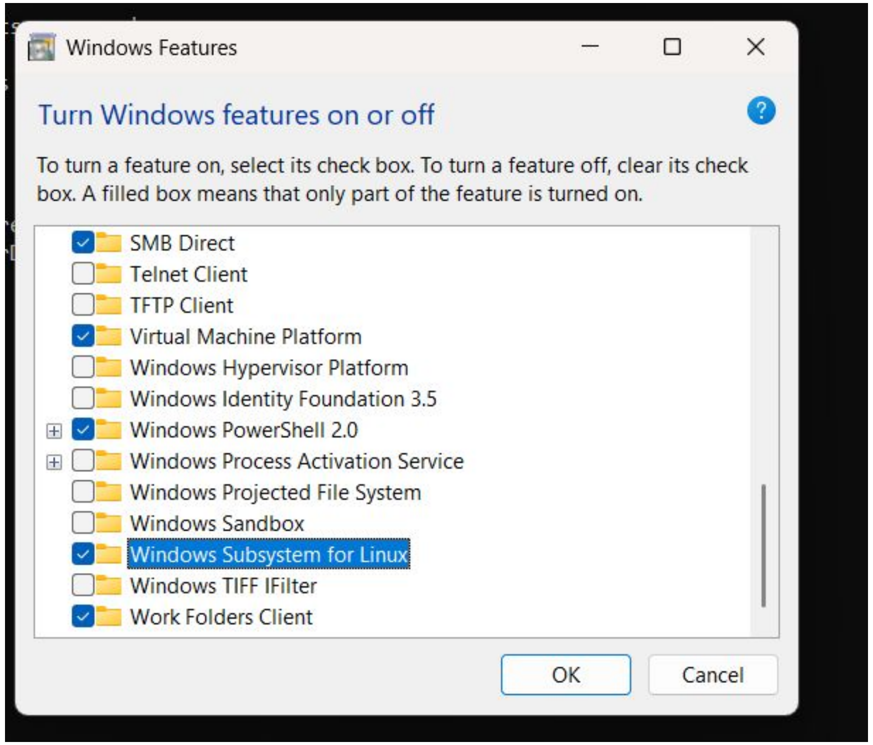
Step 1 - Open Windows Features Turn on off

Open “Turn Windows features on or off”



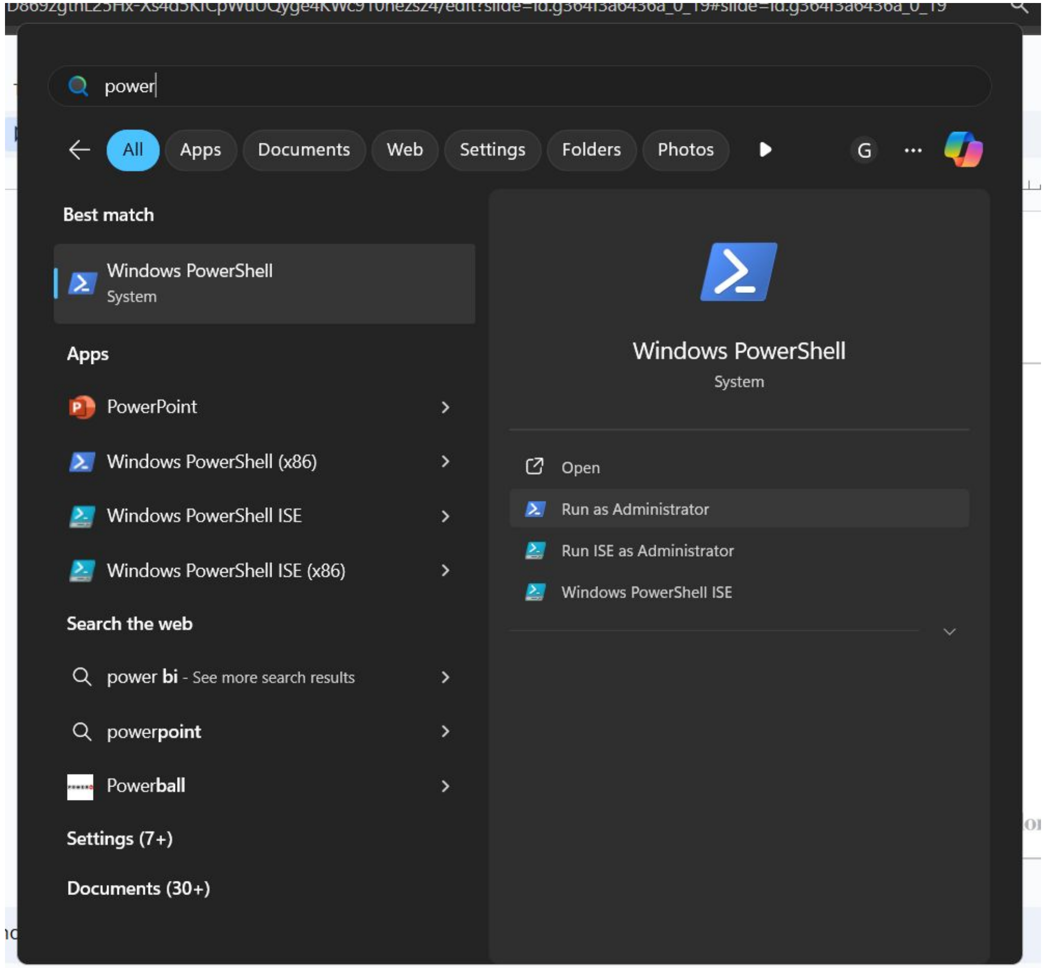
Step 2 - Enable virtualization and Linux Subsystem

Enable “Virtual Machine Platform” and “Windows Subsystem for Linux” if they were not enabled, then restart your pc



Step 3 - Install wsl

Open Powershell as administrator

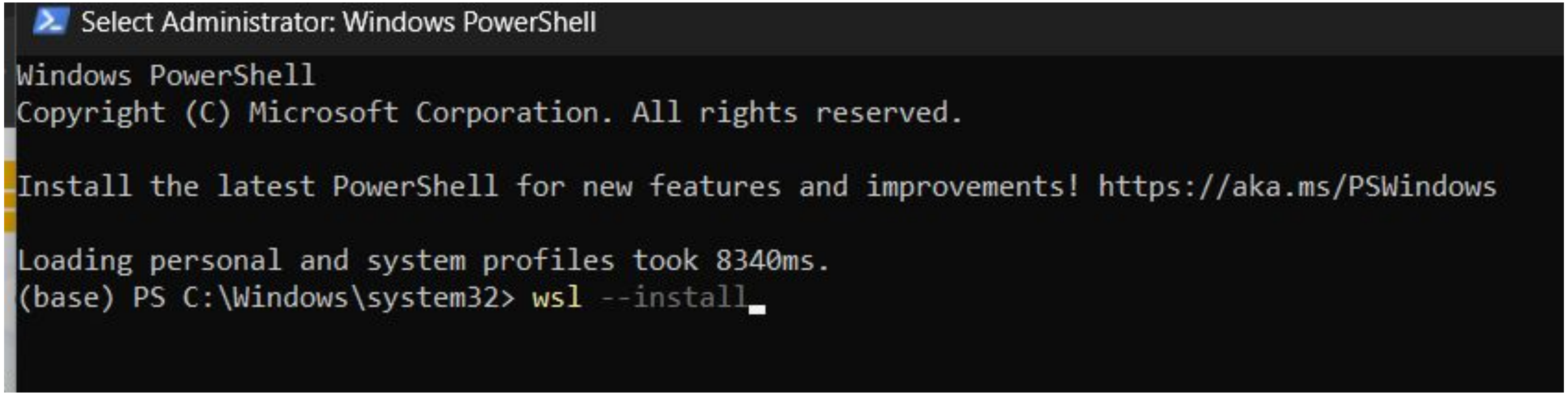


Step 4 - Install wsl

Run the following command

wsl --install

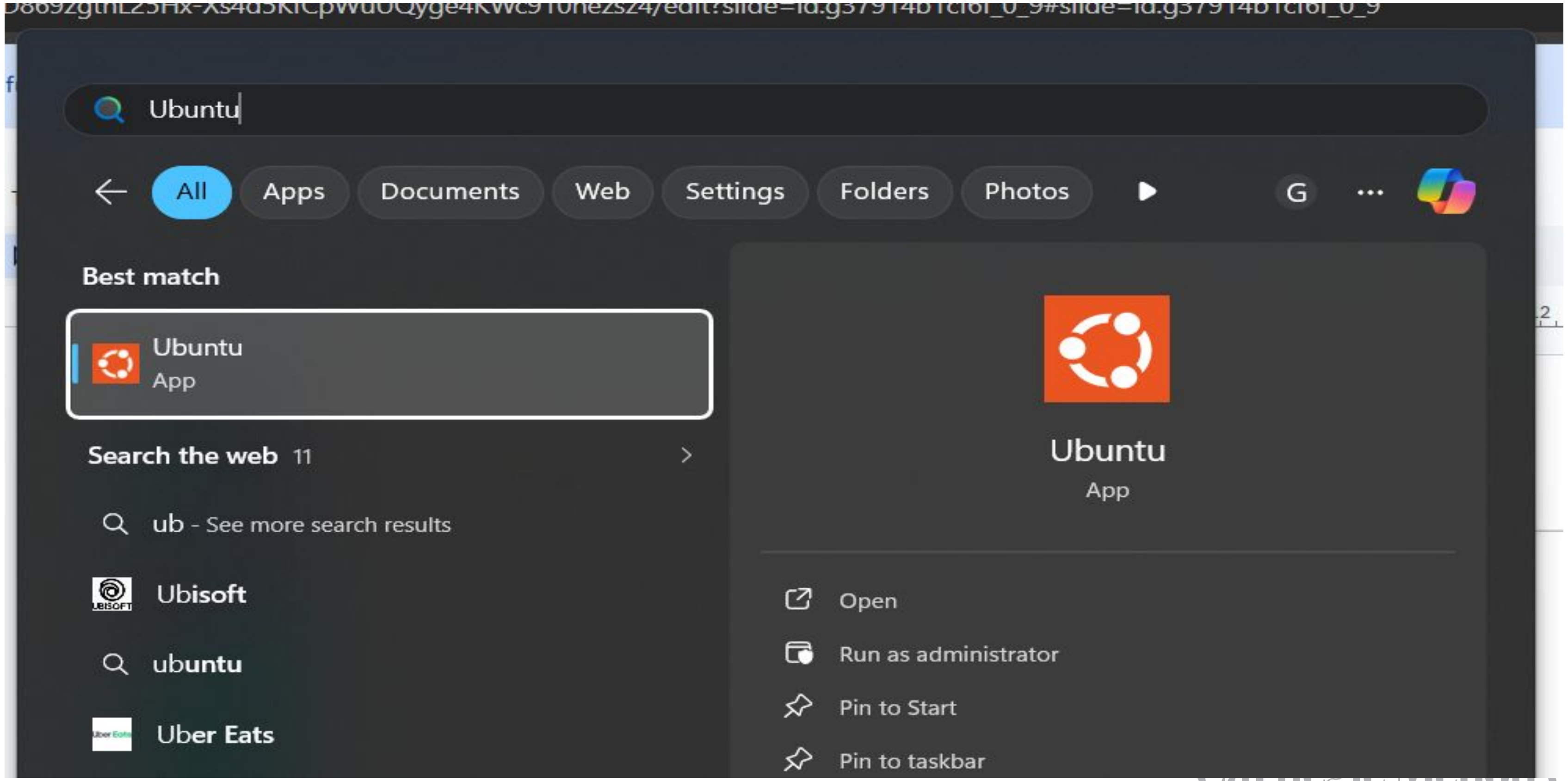
It’s two dashes



After this you will be asked to set a username and password

Step 5 - Open Ubuntu

Open it with “wsl -d Ubuntu” or do like normal people, just search “ubuntu”



Step 6 - Install packages

Run these commands

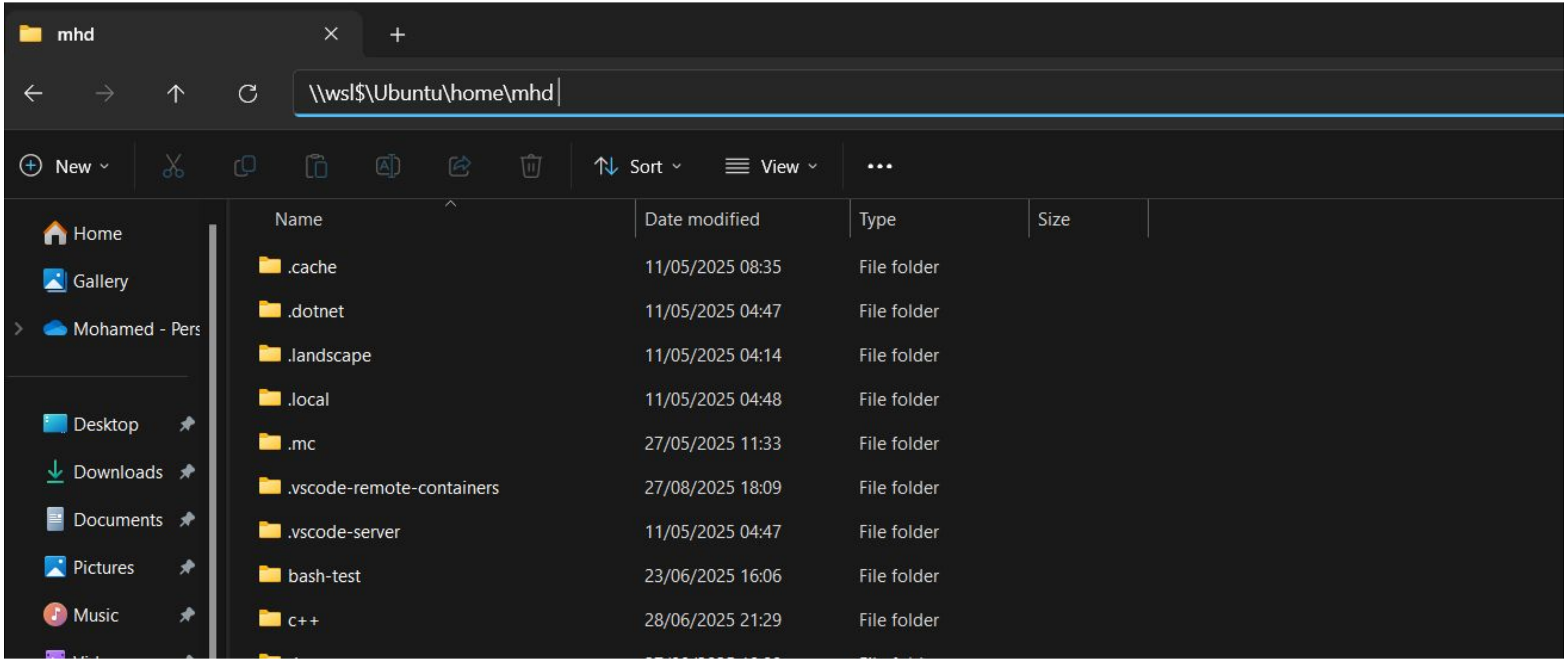
```
sudo apt update
sudo apt install -y build-essential cmake gdb pkg-config tree
sudo apt install cppcheck
sudo apt install valgrind
```

Checks

```
mhd@mhd:~$ g++ --version
g++ (Ubuntu 13.3.0-6ubuntu2~24.04) 13.3.0
Copyright (C) 2023 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

mhd@mhd:~$ gcc --version
gcc (Ubuntu 13.3.0-6ubuntu2~24.04) 13.3.0
Copyright (C) 2023 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

mhd@mhd:~$ gdb --version
GNU gdb (Ubuntu 15.0.50.20240403-0ubuntu1) 15.0.50.20240403-git
Copyright (C) 2024 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
mhd@mhd:~$ cmake --version
cmake version 3.28.3

CMake suite maintained and supported by Kitware (kitware.com/cmake).
mhd@mhd:~$ make --version
GNU Make 4.3
Built for x86_64-pc-linux-gnu
Copyright (C) 1988-2020 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
mhd@mhd:~$ cppcheck --version
Cppcheck 2.13.0
mhd@mhd:~$ valgrind --version
valgrind-3.22.0
mhd@mhd:~$
```

Browsing from the UI

You can access your home directory : \\wsl$\Ubuntu\home\<username>



Now you good, let’s set up your environment

Your folder structure should look like this, No need to create the bin folder it’s auto generated

You can place your DSA4E folder anywhere, for this tuto you can put it under ~, that will make assistance easier

```
.
└── DSA4E
    ├── assignments
    │   └── assignment0
    │       ├── CMakeLists.txt
    │       ├── bin
    │       ├── build
    │       └── src
    │           └── andrewid
    │               ├── main.cpp
    │               ├── test.cpp
    │               └── test.h
    └── labs
```

Command line to create the folders

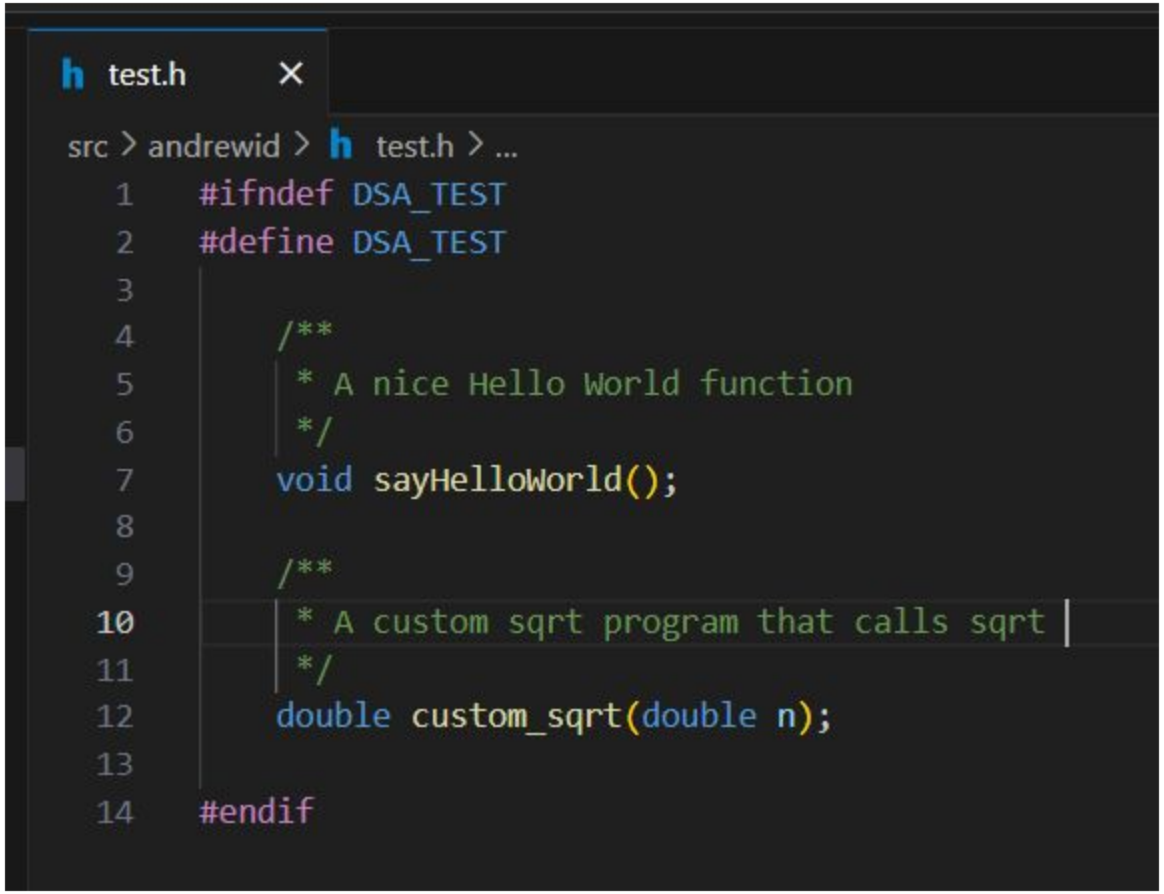
```
cd ~

mkdir -p DSA4E/{assignments/assignment0/{build,data,src/andrewid},labs}

touch DSA4E/assignments/assignment0/{CMakeLists.txt,src/andrewid/{main.cpp,test.h,test.cpp}}

tree DSA4E

cd DSA4E/assignments/assignment0/

code .
```

CMakeLists.txt content

```
cmake_minimum_required(VERSION 3.5)
project(assignment0)
file(GLOB_RECURSE assignment0_src
 "src/*.cpp"
 "src/*/*.cpp"
)
add_executable(assignment0 ${assignment0_src})
set(CMAKE_CXX_FLAGS "${CMAKE_CXX_FLAGS}  -Wall -O2")

set(CMAKE_BINARY_DIR "../bin")
set(EXECUTABLE_OUTPUT_PATH ${CMAKE_BINARY_DIR})
```

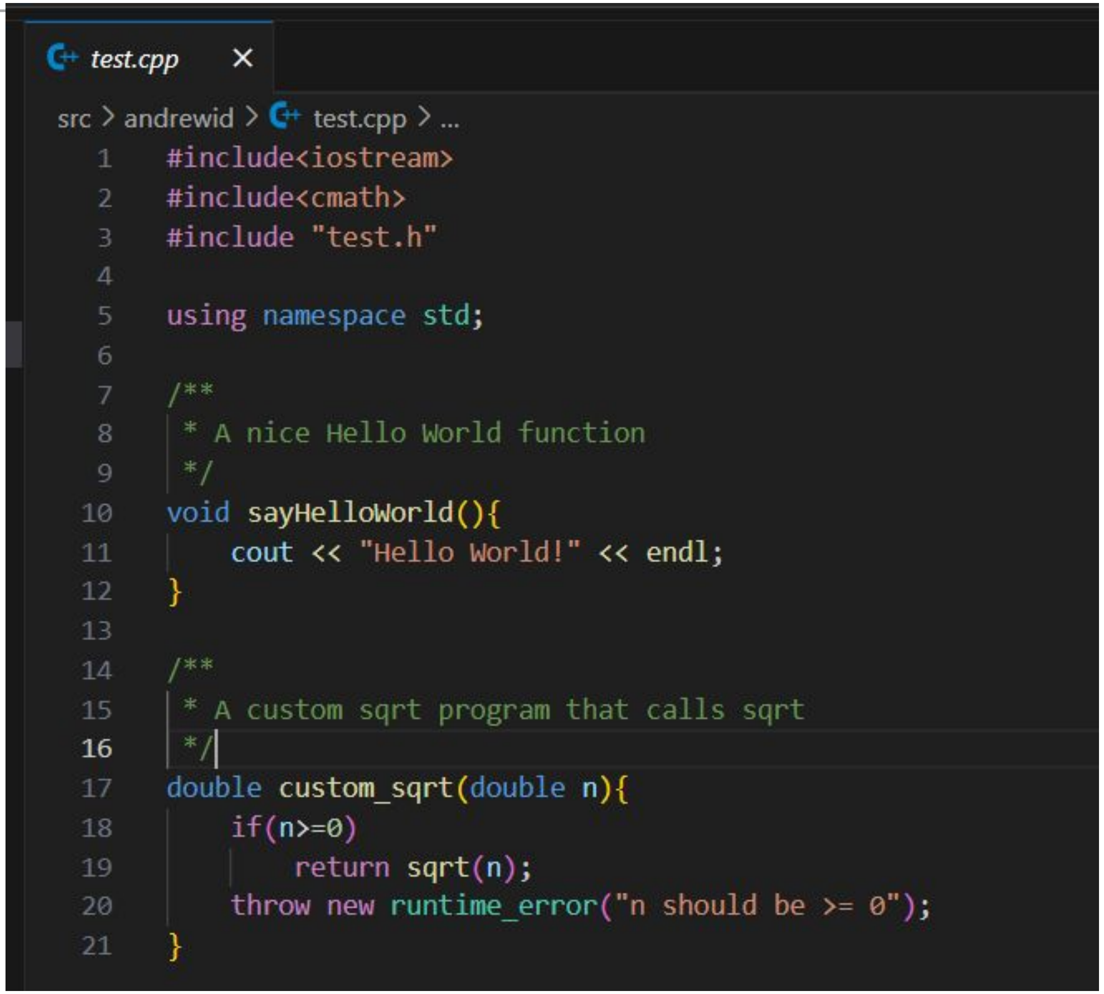
test.h content

```
#ifndef DSA_TEST
#define DSA_TEST

    /**
     * A nice Hello World function
     */
    void sayHelloWorld();

    /**
     * A custom sqrt program that calls sqrt
     */
    double custom_sqrt(double n);

#endif
```



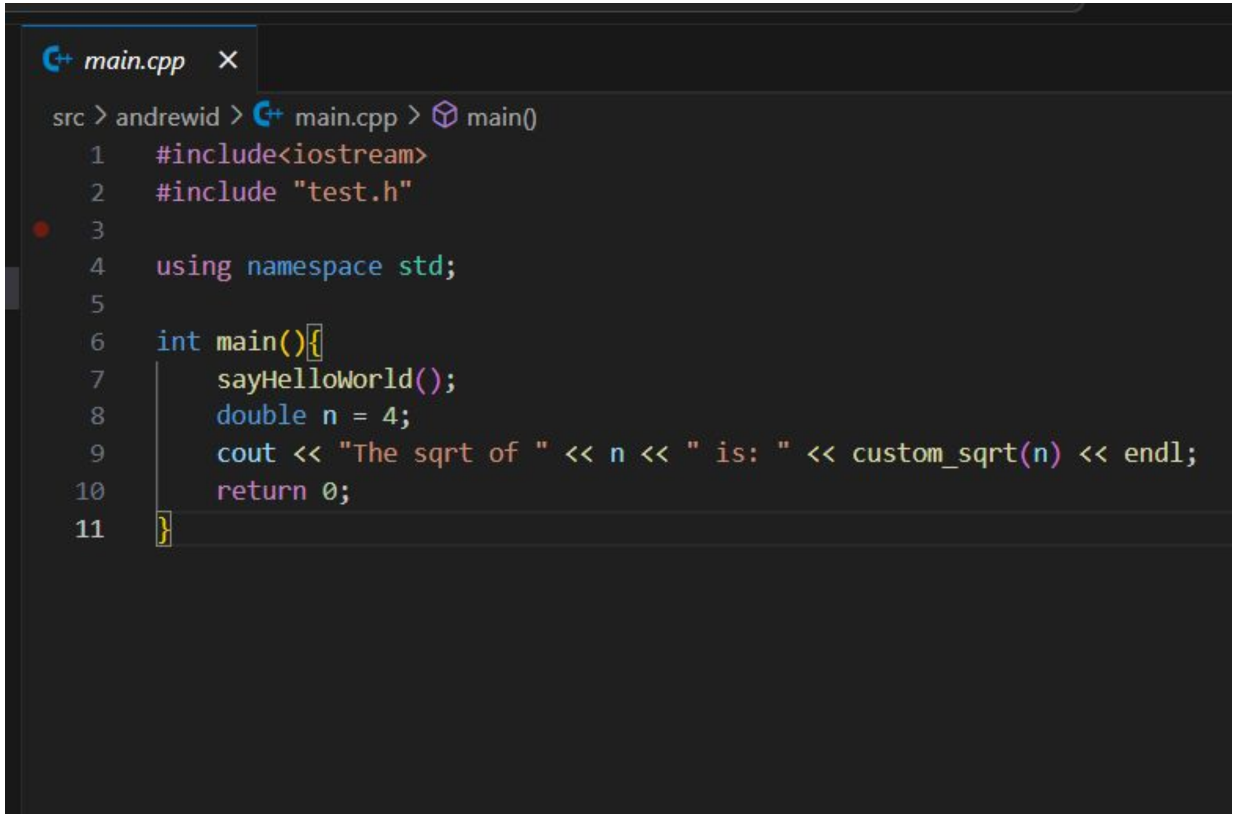
test.cpp content

```cpp
#include<iostream>
#include<cmath>
#include "test.h"

using namespace std;

/**
 * A nice Hello World function
 */
void sayHelloWorld(){
    cout << "Hello World!" << endl;
}

/**
 * A custom sqrt program that calls sqrt
 */
double custom_sqrt(double n){
    if(n>=0)
        return sqrt(n);
    throw new runtime_error("n should be >= 0");
}
```



main.cpp content

```
#include<iostream>
#include "test.h"

using namespace std;

int main(){
    sayHelloWorld();
    double n = 4;
    cout << "The sqrt of " << n << " is: " << custom_sqrt(n) << endl;
    return 0;
}
```



Compile your project

Now that you are in assignment0 folder, generate the build tools from build and build your project by running the following commands

cd build
cmake ..
make

```
mhd@mhd:~/testdsa/test2/DSA4E/assignments/assignment0$ cd build/
mhd@mhd:~/testdsa/test2/DSA4E/assignments/assignment0/build$ cmake ..
-- The C compiler identification is GNU 13.3.0
-- The CXX compiler identification is GNU 13.3.0
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Check for working C compiler: /usr/bin/cc - skipped
-- Detecting C compile features
-- Detecting C compile features - done
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: /usr/bin/c++ - skipped
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Configuring done (11.4s)
-- Generating done (0.0s)
-- Build files have been written to: /home/mhd/testdsa/test2/DSA4E/assignments/assignment0/build
mhd@mhd:~/testdsa/test2/DSA4E/assignments/assignment0/build$ make
[ 33%] Building CXX object CMakeFiles/assignment0.dir/src/andrewid/main.cpp.o
[ 66%] Building CXX object CMakeFiles/assignment0.dir/src/andrewid/test.cpp.o
[100%] Linking CXX executable /home/mhd/testdsa/test2/DSA4E/assignments/assignment0/bin/assignment0
[100%] Built target assignment0
```

Now you can execute your program from bin that's what we will do in the next slide

Run your project

Navigate to bin and run your project

cd ..
cd bin
./assignment0

```
mhd@mhd:~/DSA4E/assignments/assignment0/build$ cd ..
mhd@mhd:~/DSA4E/assignments/assignment0$ cd bin
mhd@mhd:~/DSA4E/assignments/assignment0/bin$ ./assignment0
Hello World!
The sqrt of 4 is: 2
mhd@mhd:~/DSA4E/assignments/assignment0/bin$
```

Now you good ! Enjoy C++ ;-)